

Unit Ramp Signal

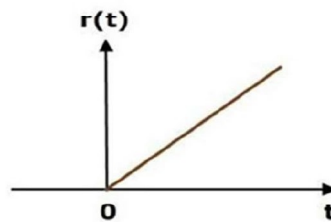
A unit ramp signal, $r(t)$ is defined as

$$r(t) = t; t \geq 0$$
$$= 0; t < 0$$

We can write unit ramp signal, $r(t)$ in terms of unit step signal, $u(t)$ as

$$r(t) = tu(t)$$

Following figure shows unit ramp signal.



The unit ramp signal is observed for all non-negative values of 't,' including zero. Its value exhibits a linear increase with respect to 't' during this interval, while the unit ramp signal remains zero for all negative values of 't'.

Unit Parabolic Signal

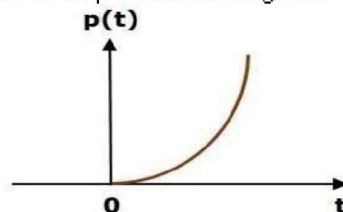
A unit parabolic signal, $p(t)$ is defined as,

$$p(t) = \frac{t^2}{2}; t \geq 0$$
$$= 0; t < 0$$

We can write unit parabolic signal, $p(t)$ in terms of the unit step signal, $u(t)$ as,

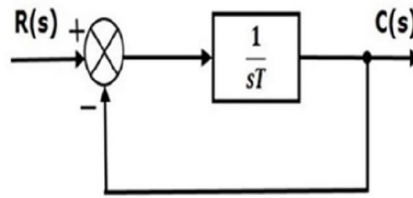
$$p(t) = \frac{t^2}{2}u(t)$$

The following figure shows the unit parabolic signal.



The unit parabolic signal is present for all non-negative values of 't,' including zero. Its value experiences a non-linear increase with respect to 't' during this interval, while the unit parabolic signal maintains a value of zero for all negative values of 't'.

In this section, we will explore the time response of a first-order system. Examine the provided block diagram of the closed-loop control system, where an open-loop transfer function of $1/s T$ is connected with unity negative feedback.



We know that the transfer function of the closed loop control system has unity negative feedback as,

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

Substitute, $G(s) = \frac{1}{sT}$ in the above equation.

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{sT}}{1 + \frac{1}{sT}} = \frac{1}{sT + 1}$$

The power of s is one in the denominator term. Hence, the above transfer function is of the first order and the system is said to be the first order system.

We can re-write the above equation as

$$C(s) = \left(\frac{1}{sT + 1} \right) R(s)$$

Where,

- **C(s)** is the Laplace transform of the output signal $c(t)$,
- **R(s)** is the Laplace transform of the input signal $r(t)$, and
- **T** is the time constant.

Follow these steps to get the response (output) of the first order system in the time domain.

- Take the Laplace transform of the input signal $r(t)$.
- Consider the equation, $C(s) = \left(\frac{1}{sT+1} \right) R(s)$
- Substitute $R(s)$ value in the above equation.
- Do partial fractions of $C(s)$ if required.
- Apply inverse Laplace transform to $C(s)$.

Impulse Response of First Order System

Consider the **unit impulse signal** as an input to the first order system.

So, $r(t) = \delta(t)$

Apply Laplace transform on both the sides.